University of North Carolina at Charlotte
Department of Electrical and Computer Engineering

Homework Report 3

ECGR 5106 - Introduction to Deep Learning

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GitHub: https://github.com/vipravlipare/ECGR-5106-Intro-To-Deep-Learning

Homework 3 Report

Problem 1: Baseline Encoder-Decoder Architecture

Objective

For this problem, a baseline sequence-to-sequence model was trained for English-to-French translation using the provided vast_english_french.txt dataset. The goal was to build a GRU encoder-decoder model, track training and validation cross-entropy loss, and evaluate the final model with exact sequence accuracy and BLEU-4.

Assumptions

The dataset was treated as a character-level translation dataset because the lecture examples used character-level recurrent models. The same random seed, seed 42, was used for all splits. The dataset contained 555 sentence pairs, with 444 pairs used for training and 111 pairs used for validation. This is a 80/20 split, and the same split membership was kept for Problems 1, 2, and 3.

Model Setup

The baseline model used one GRU encoder and one GRU decoder. The model was character-level, so each sentence was converted into individual characters instead of full words. Special tokens were added for padding, start of sentence, and end of sentence. The vocabulary had 48 total characters. The encoder used an embedding layer, dropout, and one GRU layer with hidden_size = 160. The decoder used the same hidden size, then used a fully connected layer to predict the next French character. The encoder had 162,240 parameters and the decoder had 169,968 parameters, for a total of 332,208 trainable parameters. This model was chosen because it is simple and works as a good baseline before adding attention. The main weakness is that the encoder has to store the whole English sentence in one hidden state, so longer sentences can lose information.

Training Setup and Hyperparameters

The dataset had 555 total English-French sentence pairs. A random seed of 42 was used. The data was split into 444 training pairs and 111 validation pairs, which is the required 80/20 split. The model was trained with Adam, learning rate = 0.025, weight decay = 5e-5, batch size = 96, dropout = 0.35, and NLLLoss. The PAD token was ignored in the loss so padding would not affect training. ReduceLROnPlateau was used to lower the learning rate when validation loss stopped improving. Early stopping was also used. The model was allowed to train up to 500 epochs, but it stopped after 70 epochs. The best validation loss happened at epoch 44. Training was done on CPU and took about 61.77 seconds.

Problem 1 Results

The traditional sequence accuracy was 0.00%, so none of the validation translations matched the target sentence perfectly. This metric is very strict because the whole sentence has to be exactly correct. One wrong word, missing accent, or extra word makes the answer count as wrong.

The word-level BLEU-4 score was 0.0501. This was low because the predicted sentences usually did not match many full French words from the target. The character-level BLEU-4 was higher at 0.1881 because the model was trained character by character. This shows that the model learned some French-looking spelling and sentence patterns, but it did not usually generate the full correct translation.

The final training loss was 0.4441, while the final validation loss was 1.4277. The best validation loss was 1.4115 at epoch 44. Since the training loss was much lower than the validation loss, the model was learning the training data better than the validation data. This shows some overfitting.

Problem 1 result table

Model	Direction	Split	Epochs	Exact Acc	Word BLEU-4	Char BLEU-4	Best Val Loss	Params
Baseline GRU Encoder-Decorder	English to French	80/20	70	0.00%	0.0501	0.1881	1.4115	332,208

Qualitative Validation

The baseline model learned some French-looking character patterns, but most predictions were still not close enough to match the target sentences. This is consistent with the low exact accuracy and low word-level BLEU score.

The sample translations show that the model learned some useful patterns, but still made many mistakes. For example, it often started with reasonable French words like “je,” “il,” “elle,” or “nous,” but then the rest of the sentence drifted away from the correct target. One stronger example was “i want a hot cup of coffee,” where the model predicted a sentence close to “je veux une tasse de chaud,” but it still missed the word for coffee.

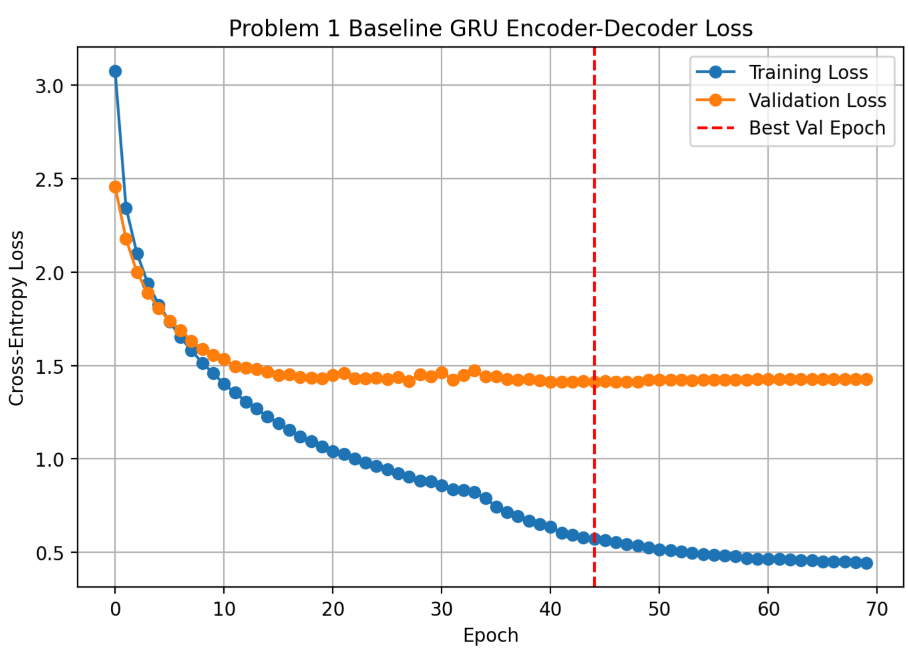
Overall, the baseline GRU learned basic French character patterns, but it was not strong enough to remember the full sentence meaning. This makes sense because the whole English sentence is compressed into one hidden state. This is why attention is useful to test in Problem 2.

English Input	Target French	Predicted French	Exact - Result	Word BLEU-4	Char BLEU-4
They feed the	ils	ils vont à la	Fail	0.0285	0.2335

pigeons in the square	nourrissent les pigeons sur la place	porte a une tournoi prochaine			
she practices yoga every morning	elle pratique le yoga tous les matins	elle a perdu matin est très compétitif	Fail	0.0330	0.2991
i enjoy walking in the snow	j'aime marcher dans la neige	j'ai fermement le couramment	Fail	0.0000	0.1545
she loves to wear modern jackets	elle adore porter des vestes modernes	elle a perdu ses clés de collection des re...	Fail	0.0157	0.1701
she loves to wear modern jackets	le bus de la ville arrive précisément à ci...	le chien attrape la biologique	Fail	0.02335	0.1166

Problem 1 Plots

The loss curve shows the baseline training loss decreasing much more clearly than the validation loss. This means the model learned the training pairs, but the single hidden vector bottleneck made it hard to generalize to unseen validation sentences. Since the decoder only receives the final encoder hidden state, it has to store the whole source sentence in one compressed representation. The training and validation losses would stay the same level until around 1.4 - 1.5, and after that the model would start overfitting.



Problem 2: Sequence-to-Sequence with Attention

Objective

For Problem 2, the baseline encoder-decoder was extended with an attention mechanism. The purpose was to test whether allowing the decoder to look back at encoder states during generation improved validation loss, exact sequence accuracy, and BLEU-4 compared with the baseline model.

Model Setup

The attention model used the same basic encoder-decoder idea as Problem 1, but it added an attention layer. In Problem 1, the decoder mainly depended on one final hidden state from the encoder. In Problem 2, the encoder returned all hidden states, so the decoder could look back at different parts of the input sentence while generating each French character. The model used Luong-style attention. This means that at each decoding step, the decoder gave different weights to different source characters. The encoder was also bidirectional, so it read the English sentence forward and backward. This helped the model understand more context from the input sentence. Beam search with beam size 3 was used during prediction. Instead of only picking the best character at each step, beam search kept a few possible translations and chose the best overall one. This model was larger than the baseline, but it was designed to fix the main weakness from Problem 1, where the whole sentence had to fit into one hidden state.

Training Setup and Hyperparameters

The same 80/20 train-validation split from Problem 1 was used, so the results could be compared fairly. The model used hidden_size = 160, dropout = 0.08, Adam optimizer, learning rate = 0.003, weight decay = 1e-5, batch size = 96, and beam size = 3. The maximum epoch count was 95, and the final saved run trained for 65 epochs. The model was trained on CPU. The best checkpoint was chosen using sampled BLEU and similarity instead of only validation loss. This was done because the lowest validation loss did not always give the best generated translations.

Problem 2 Results

The attention model improved the translation results compared to the baseline model. Exact sequence accuracy increased from 0.00% to 0.90%. This is still low, but it means the attention model was able to produce at least one perfect validation translation, while the baseline produced none. The word-level BLEU-4 score improved from 0.0501 to 0.0899. The character-level BLEU-4 score also improved from 0.1881 to 0.2620. This shows that attention helped the model generate translations that were closer to the target sentences, even when the full sentence was not exactly correct. The main result is that attention helped more with partial translation quality than exact-match accuracy. Exact-match accuracy stayed low because the metric is very strict, but BLEU improved because the attention model matched more words and character patterns from the correct French sentences.

Problem 2 result table

Model	Exact Acc	Word BLEU-4	Char BLEU-4	Best Val Loss	Time (s)	Params
Baseline GRU	0.00%	0.0501	0.1881	1.4115	61.77	332,208
Attention GRU	0.90%	0.0899	0.2620	1.4355	259.55	666,929

Qualitative Validation

The attention model produced better translations than the baseline because it could look back at different parts of the English sentence while generating the French output. The attention maps also support this because they show the decoder focusing on different source characters instead of using only one final hidden state.

The first table shows the regular validation examples used in the report. These examples are not exact matches, but some of them still have decent BLEU scores because parts of the sentence are close to the target.

English Input	Target French	Predicted French	Exact - Result	Word BLEU-4	Char BLEU-4
they feed the pigeons in the square	ils nourrissent les pigeons sur la place	ils nourrissent les excentre été	Fail	0.1782	0.5103
she practices yoga every morning	elle pratique le yoga tous les matins	elle joue d'un chaque jour	Fail	0.0360	0.1879
i enjoy walking in the snow	j'aime marcher dans la neige	j'aime marcher dans la pluie	Fail	0.6687	0.8280
she loves to wear modern jackets	elle adore porter des vestes modernes	elle adore porter des conserdi	Fail	0.5475	0.6116
the city bus arrives precisely at five o'c...	le bus de la ville arrive précisément à ci...	le bus arrive à cinq heurse	Fail	0.0663	0.3590

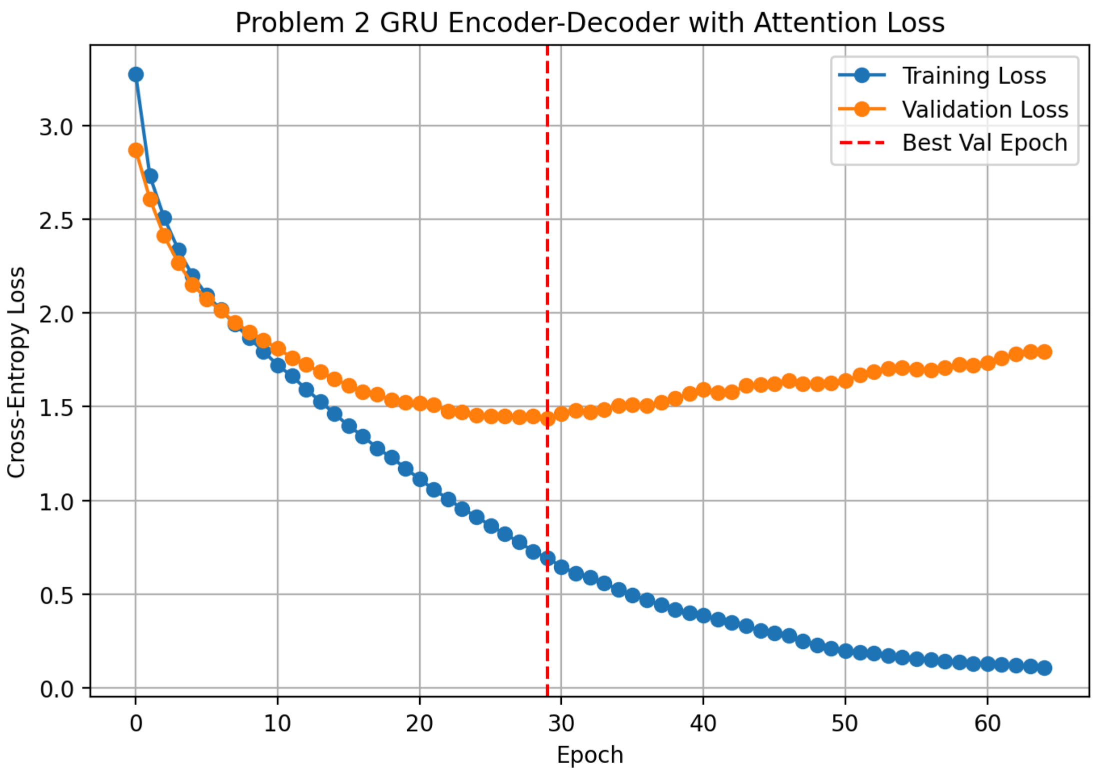
The second table shows the strongest validation examples. The first row is the one exact-match sentence found by the attention model. The other two rows are not perfect, but they still scored high because most of the sentence structure matched the target.

Overall, attention improved the model, but it did not completely solve the translation problem. It gave better BLEU scores, better character-level matching, and slightly better exact-match accuracy compared to Problem 1.

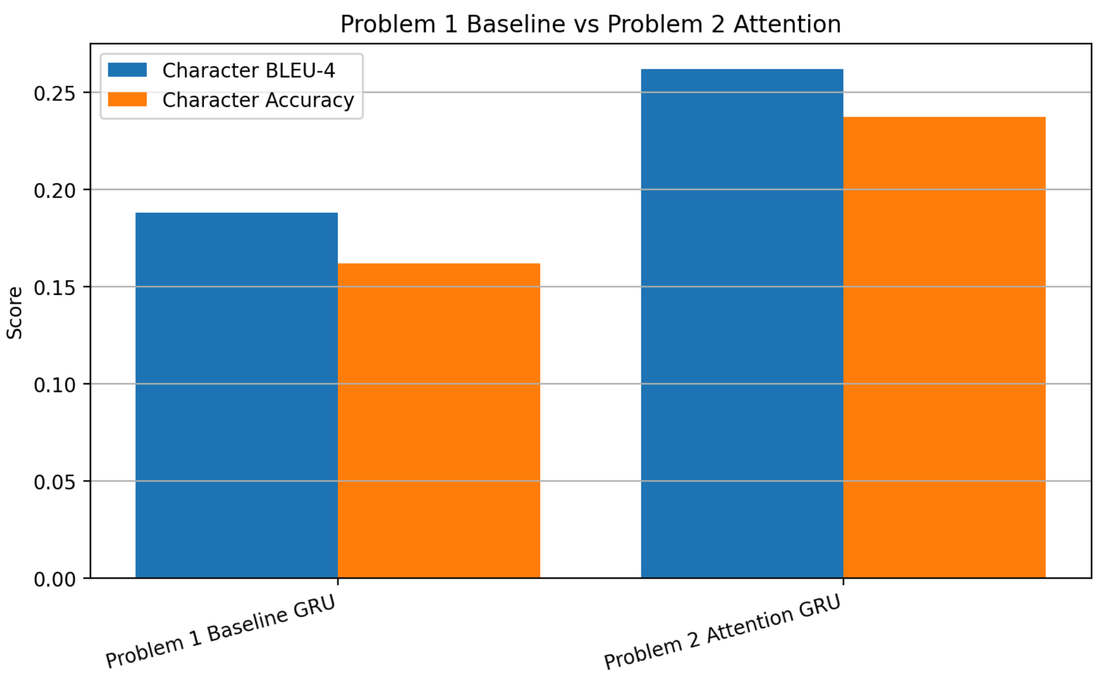
English Input	Target French	Predicted French	Exact - Result	Word BLEU-4	Char BLEU-4
i enjoy walking in the forest	j'aime marcher dans la forêt	j'aime marcher dans la forêt	Pass	1.0000	1.0000
we are planning a trip to rome	nous planifions un voyage à rome	nous planifions un voyage à l'été	Fail	0.7598	0.8411
i enjoy walking in the snow	j'aime marcher dans la neige	j'aime marcher dans la pluie	Fail	0.6687	0.8280

Problem 2 Plots and Attention Maps

The Problem 2 loss curve shows that the model fit the training data strongly, while validation loss stopped improving after a certain point. This means attention improved translation quality, but the dataset was still small enough for overfitting to happen. The comparison plot shows that attention improved BLEU and character accuracy, but the exact-match metric remained difficult because exact sentence matching gives no partial credit. Similar to problem 1, the training and validation losses would stay the same level until around 1.4 - 1.5, and after that the model would start overfitting.

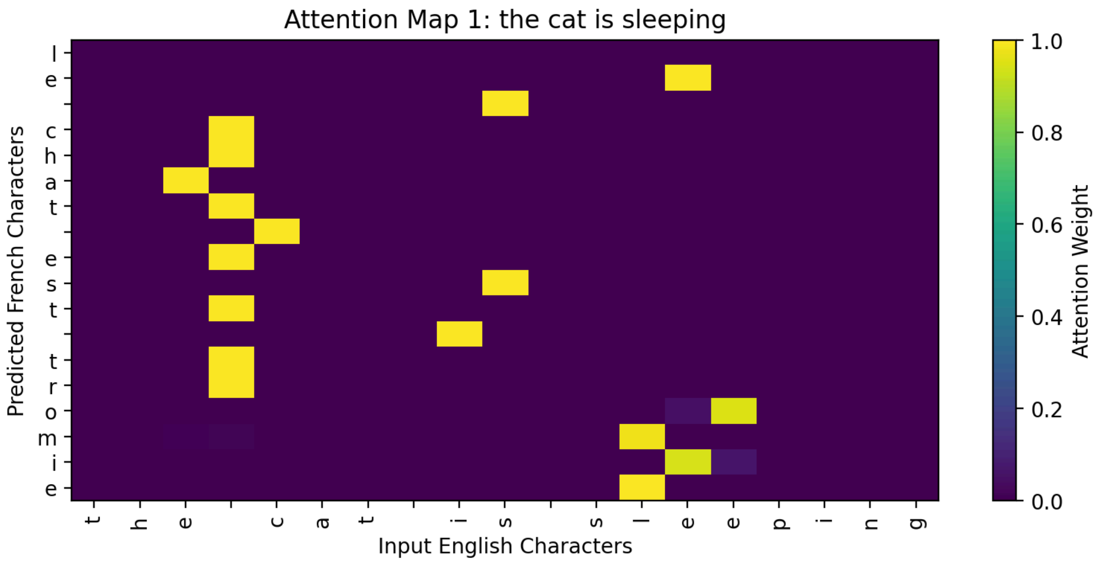


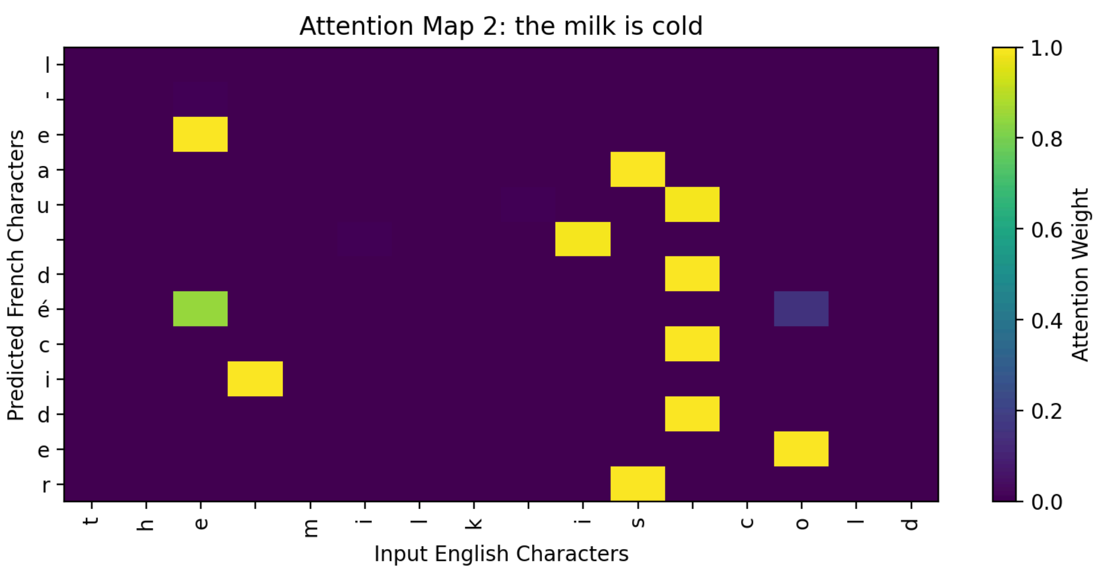
Problem 2 GRU Encoder-Decoder with Attention Loss
Training Loss
Validation Loss
Best Val Epoch
Cross-Entropy Loss
Epoch
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
10
20
30
40
50
60



Problem 1 Baseline vs Problem 2 Attention
Character BLEU-4
Character Accuracy
Score
0.00
0.05
0.10
0.15
0.20
0.25
Problem 1 Baseline GRU
Problem 2 Attention GRU

The attention maps show which source characters the decoder used while producing each output character. The alignments are not perfect, but they show that the decoder is no longer forced to depend only on one final encoder hidden state. Some target characters attend to the matching region of the English input, while other positions are more spread out because character-level translation has flexible word order and spelling differences.





Problem 3: Reversing the Language Direction

Objective

For Problem 3, the translation direction was reversed from French-to-English. The same split membership from the earlier problems was used, but the source and target languages were swapped. Both the baseline GRU model and attention GRU model were trained and evaluated in the reversed direction.

Model Setup

The reversed baseline model used the same GRU encoder-decoder structure from Problem 1, but the translation direction was changed. Instead of English to French, the input was French and the target output was English.

The reversed attention model used the same attention structure from Problem 2. It still used a bidirectional encoder, attention weights, and beam search. This made the comparison fair because the main change was the translation direction, not the model type.

The baseline model had 332,689 trainable parameters. The attention model had 666,929 trainable parameters. The attention model was larger, but it was expected to perform better because it could look back at different parts of the French input while generating the English output.

Training Setup and Hyperparameters

The same 80/20 split was used again, with 444 training sentences and 111 validation sentences. The only difference was that the source and target languages were reversed.

The reversed baseline used hidden_size = 160, learning rate = 0.010, dropout = 0.20, weight decay = 0.00005, and beam size = 3. It trained for 58 epochs on CPU and took about 202.59 seconds.

The reversed attention model used hidden_size = 160, learning rate = 0.003, dropout = 0.08, weight decay = 0.00001, and beam size = 3. It trained for 67 epochs on CPU and took about 527.80 seconds. Both models used the best sampled BLEU/similarity checkpoint because the best validation loss did not always give the best generated translations.

Problem 3 Results

For the reversed French-to-English task, the attention model performed better than the reversed baseline. The reversed baseline had a word-level BLEU-4 score of 0.0623, while the reversed attention model improved to 0.0726. The character-level BLEU-4 also improved from 0.1711 to 0.1940.

The attention model also had better character accuracy and sequence similarity. Character accuracy improved from 18.13% to 20.83%, and sequence similarity improved from 39.72% to 44.35%. Exact sequence accuracy was 0.00% for both models, which shows that exact match was still too strict for this dataset.

Problem 3 result table

Model	Exact Acc	Word BLEU-4	Char BLEU-4	Similarity	Best Val Loss	Time (s)	Params
Reversed Baseline GRU	0.00%	0.0623	0.1711	0.3972	1.3773	202.59	332,689
Reversed Attention GRU	0.00%	0.0726	0.1940	0.4435	1.4748	527.80	666,929

Qualitative Validation

The results show that attention helped in the reversed direction too, but the improvement was smaller than in Problem 2. The attention model produced translations that were closer to the target English sentences, but it still did not produce perfect full-sentence matches.

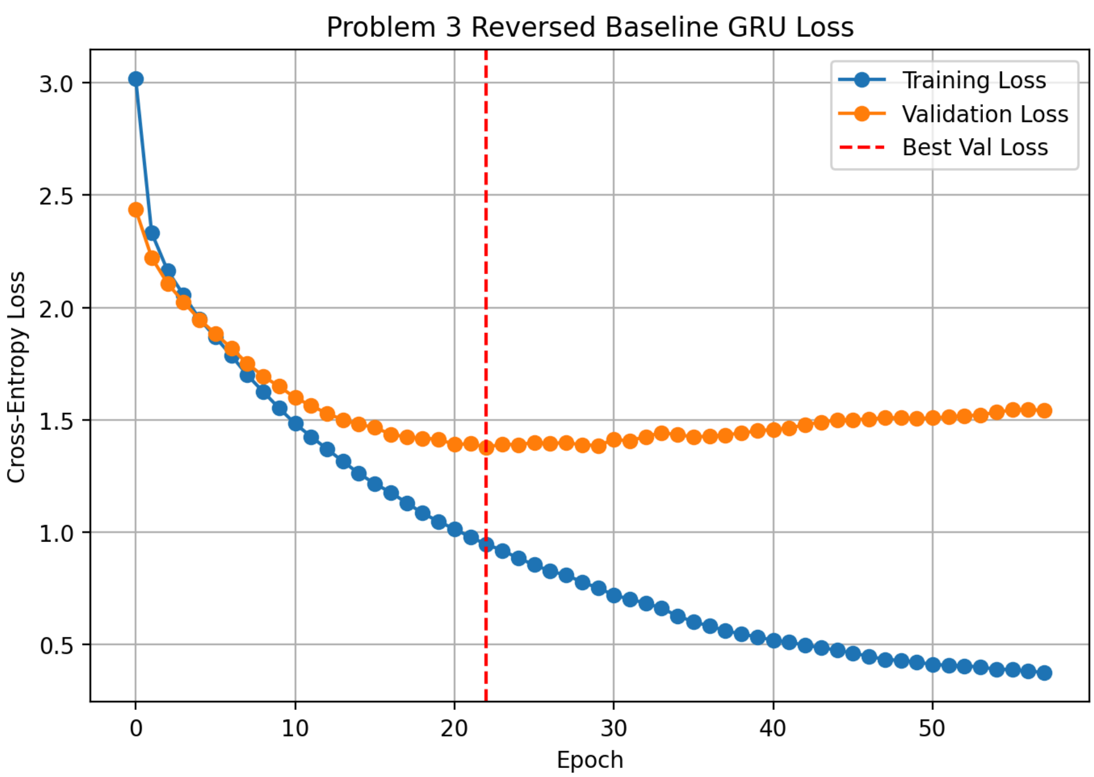
Compared to English-to-French, the French-to-English direction was not stronger in this experiment. The best English-to-French attention model had a word BLEU-4 of 0.0899 and character BLEU-4 of 0.2620. The best French-to-English attention model had a lower word BLEU-4 of 0.0726 and character BLEU-4 of 0.1940. Based on these results, English-to-French worked better overall for this dataset split.

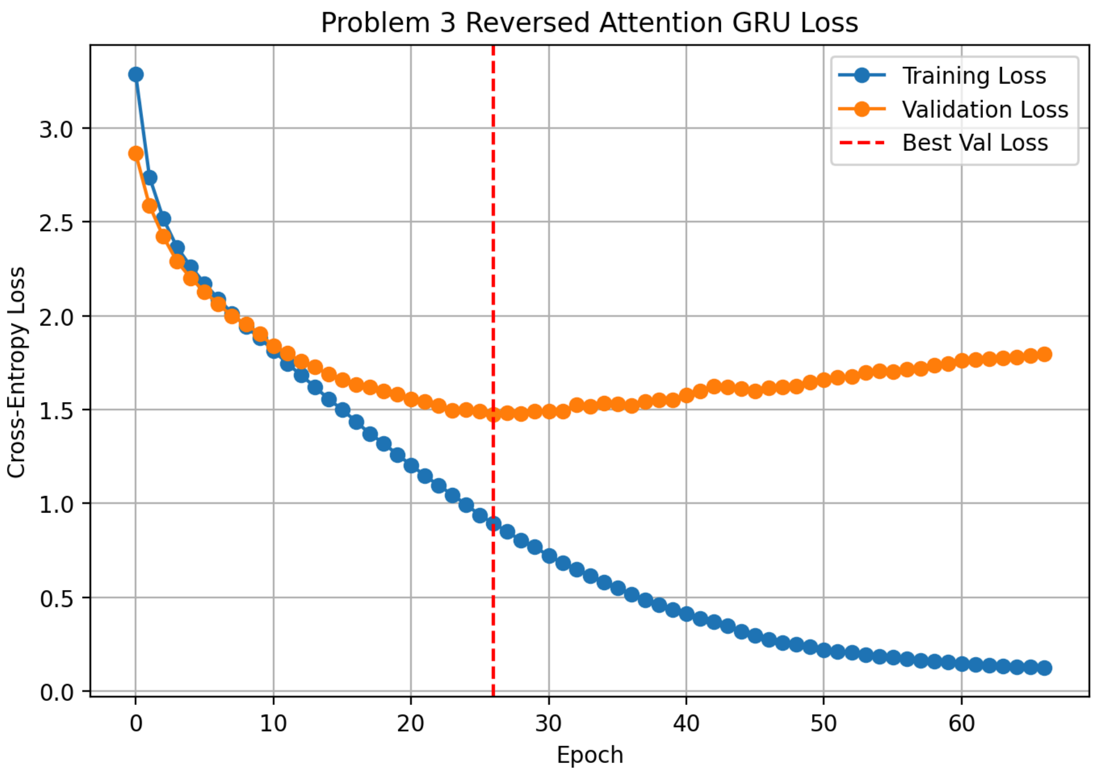
French Input	Target English	Baseline Pred	Base Exact Results	Base Word BLEE-4	Attention Pred	Attention Exact Results	Attention Word BLEU-4
ils nourrissent les pigeons sur...	they feed the pigeons in the sq...	they play tennis on the weekends	Fail	0.0411	they feed the park	Fail	0.1878
elle pratique le yoga tous les ...	she practices yoga every morning	she translates legal agreements	Fail	0.0626	she play long	Fail	0.0583
j'aime marcher dans la neige	i enjoy walking in the snow	i look forward to the party	Fail	0.0485	i look for holiday	Fail	0.0487
elle adore porter des	she loves to wear modern jackets	she loves to wear long skirts	Fail	0.5081	she loves to works	Fail	0.2412

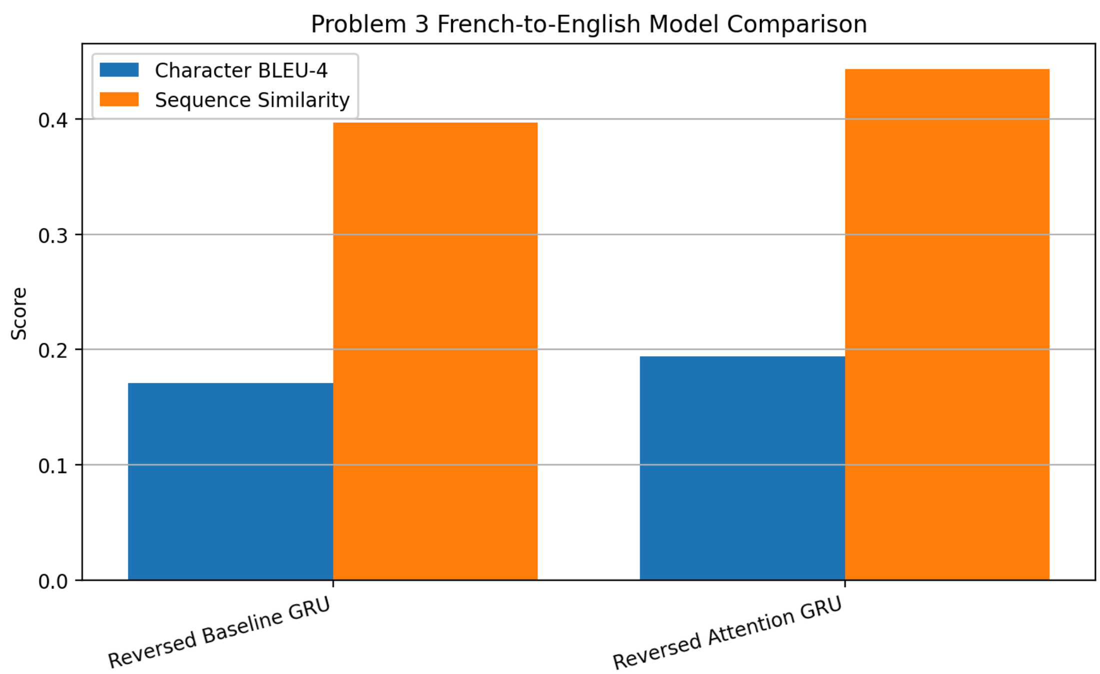
vestes mo...							
le bus de la ville arrive préci...	the city bus arrives precisely ...	the bread smells wonderful ever...	Fail	0.0286	the fresh conferen ce	Fail	0.0215

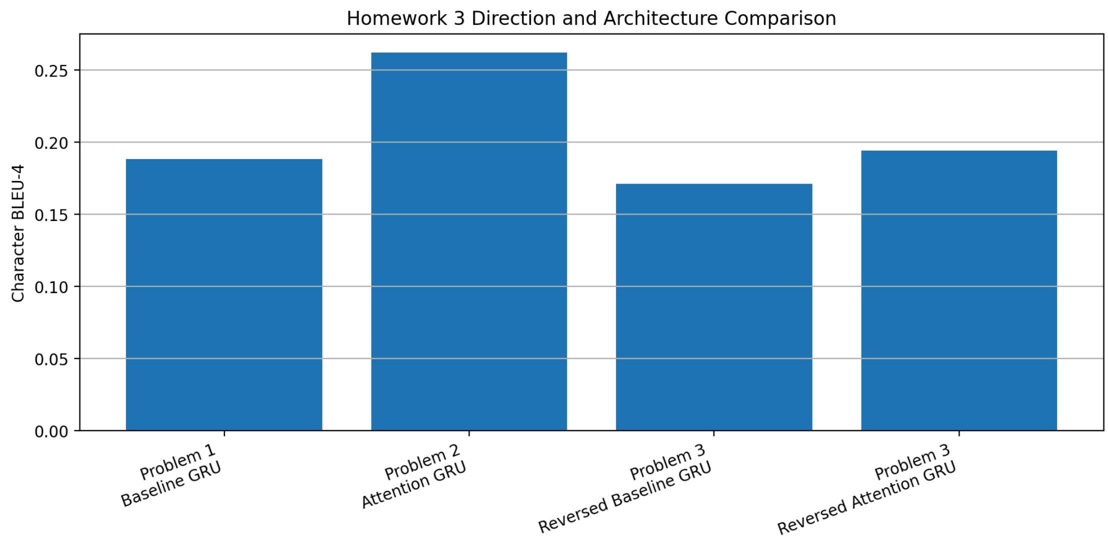
Problem 3 Plots

The reversed loss curves show the same overall pattern as the English-to-French models: training loss drops faster than validation loss. The attention model used more parameters and took longer to train, but it gave better BLEU and similarity. The model comparison plot shows that the reversed attention model was the best reversed-direction model, while the full BLEU comparison shows that the English-to-French attention model from Problem 2 was still the strongest overall run.

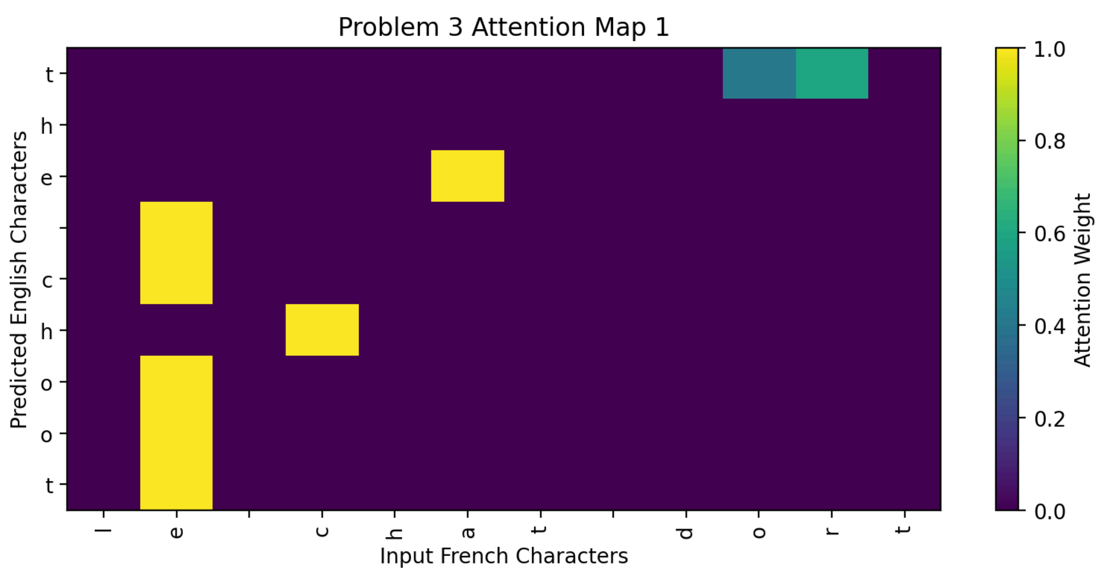


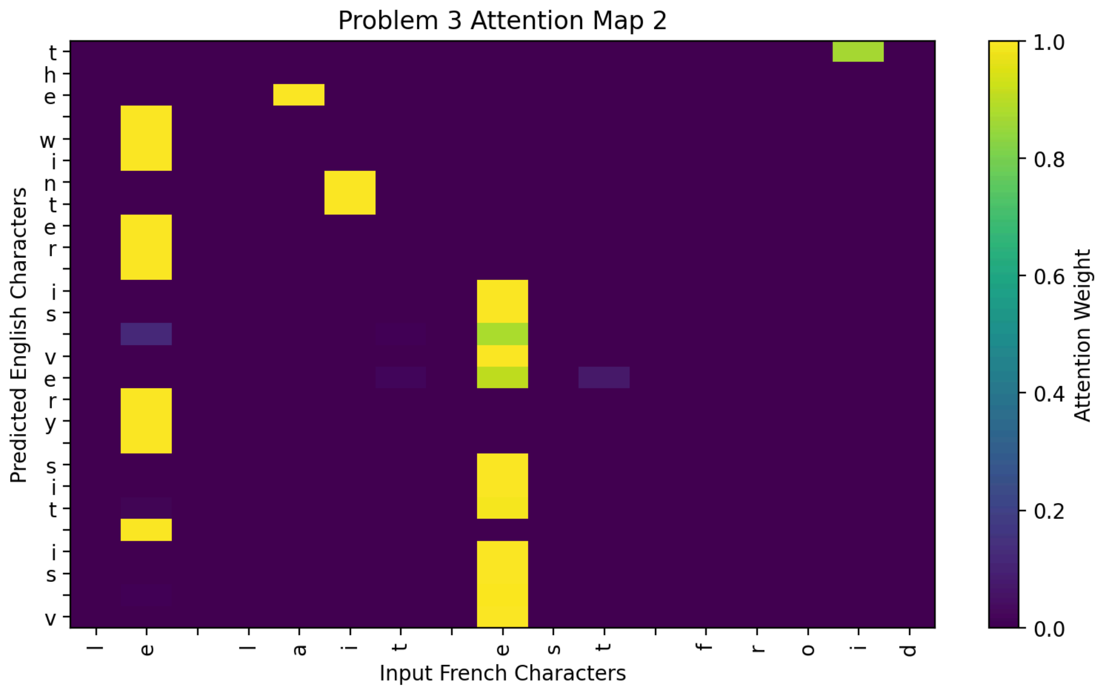

Problem 3 Reversed Attention GRU Loss
Training Loss
Validation Loss
Best Val Loss
Cross-Entropy Loss
Epoch
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
10
20
30
40
50
60


Problem 3 French-to-English Model Comparison
Character BLEU-4
Sequence Similarity
Score
0.0
0.1
0.2
0.3
0.4
Reversed Baseline GRU
Reversed Attention GRU



The reversed attention maps show how the French input characters were weighted while generating English output. The maps are useful because they show that attention is trying to line up parts of the input and output instead of treating the whole source sentence as one vector. The alignments are still imperfect because the dataset is small and the model is character-level.





Overall Comparison and Conclusion

Across all four final runs, the attention model in the original English-to-French direction performed best. It had the highest exact sequence accuracy at 0.90%, the highest word-level BLEU-4 at 0.0899, the highest character-level BLEU-4 at 0.2620, and the strongest sequence similarity. The baseline English-to-French model was fastest, but it had the weakest qualitative translations. Adding attention roughly doubled the parameter count, but it improved the translation outputs because the decoder could look back at the encoder states instead of depending only on one final hidden vector.

Final best-result summary

Problem	Model	Direction	Exact Acc	Word BLEU-4	Char BLEU-4	Best Val Loss	Time (s)	Params
Problem 1	Baseline GRU	English to French	0.00%	0.0501	0.1881	1.4115	61.77	332,208
Problem 2	Attention GRU	English to French	0.90%	0.0899	0.260	1.4355	259.55	666,929
Problem 3	Reverse Baseline GRU	French to English	0.00%	0.0623	0.1711	1.3773	202.59	332,689

Problem 3	Reverse Attention GRU	French to English	0.00%	0.0726	0.1940	1.4748	527.80	666,929

The French-to-English direction was not easier in these results. The reversed attention model improved over the reversed baseline, but it did not beat the English-to-French attention model. The best English-to-French attention model reached 0.2620 character BLEU-4 and 0.0899 word BLEU-4, while the reversed attention model reached 0.1940 character BLEU-4 and 0.0726 word BLEU-4. Exact accuracy stayed low for all models because a sentence must be perfectly correct to count. BLEU and similarity give a more realistic view here because they show partial credit for outputs that share words, phrases, or character patterns with the target sentence.

Overall, attention was worth the extra cost for this assignment. It trained slower and used more parameters, but it improved translation quality in both directions. The main limitation was the small dataset and character-level modeling setup. A larger dataset, word/subword tokenization, and more training time would likely improve exact translation quality, but for this homework the attention GRU English-to-French model was the best final model.

All source code, notebooks, plots, and result CSV files are included in the Homework_3 folder of the GitHub repository.